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Section 12.7 : Calculus With Vector Functions

In this section we need to talk briefly about limits, derivatives and integrals of vector functions. As you will see, these behave in a fairly predictable manner. We will be doing all of the work in $\mathbb{R}^3$ but we can naturally extend the formulas/work in this section to $\mathbb{R}^n$ (*i.e.* n -dimensional space).

Let's start with limits. Here is the limit of a vector function.

$$\begin{aligned}\lim_{t \rightarrow a} \vec{r}(t) &= \lim_{t \rightarrow a} \langle f(t), g(t), h(t) \rangle \\ &= \left\langle \lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t) \right\rangle \\ &= \lim_{t \rightarrow a} f(t) \vec{i} + \lim_{t \rightarrow a} g(t) \vec{j} + \lim_{t \rightarrow a} h(t) \vec{k}\end{aligned}$$

So, all that we do is take the limit of each of the component's functions and leave it as a vector.

Example 1 Compute $\lim_{t \rightarrow 1} \vec{r}(t)$ where $\vec{r}(t) = \left\langle t^3, \frac{\sin(3t - 3)}{t - 1}, e^{2t} \right\rangle$.

Hide Solution ▼

There really isn't all that much to do here.

$$\begin{aligned}
 \lim_{t \rightarrow 1} \vec{r}(t) &= \left\langle \lim_{t \rightarrow 1} t^3, \lim_{t \rightarrow 1} \frac{\sin(3t - 3)}{t - 1}, \lim_{t \rightarrow 1} e^{2t} \right\rangle \\
 &= \left\langle \lim_{t \rightarrow 1} t^3, \lim_{t \rightarrow 1} \frac{3 \cos(3t - 3)}{1}, \lim_{t \rightarrow 1} e^{2t} \right\rangle \\
 &= \langle 1, 3, e^2 \rangle
 \end{aligned}$$

Notice that we had to use **L'Hospital's Rule** on the y component.

Now let's take care of derivatives and after seeing how limits work it shouldn't be too surprising that we have the following for derivatives.

$$\vec{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle = f'(t) \vec{i} + g'(t) \vec{j} + h'(t) \vec{k}$$

Example 2 Compute $\vec{r}'(t)$ for $\vec{r}(t) = t^6 \vec{i} + \sin(2t) \vec{j} - \ln(t + 1) \vec{k}$.

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There really isn't too much to this problem other than taking the derivatives.

$$\vec{r}'(t) = 6t^5 \vec{i} + 2 \cos(2t) \vec{j} - \frac{1}{t + 1} \vec{k}$$

Most of the basic facts that we know about derivatives still hold however, just to make it clear here are some facts about derivatives of vector functions.

Facts

$$\begin{aligned}\frac{d}{dt}(\vec{u} + \vec{v}) &= \vec{u}' + \vec{v}' \\ (c\vec{u})' &= c\vec{u}' \\ \frac{d}{dt}(f(t)\vec{u}(t)) &= f'(t)\vec{u}(t) + f(t)\vec{u}'(t) \\ \frac{d}{dt}(\vec{u} \cdot \vec{v}) &= \vec{u}' \cdot \vec{v} + \vec{u} \cdot \vec{v}' \\ \frac{d}{dt}(\vec{u} \times \vec{v}) &= \vec{u}' \times \vec{v} + \vec{u} \times \vec{v}' \\ \frac{d}{dt}(\vec{u}(f(t))) &= f'(t)\vec{u}'(f(t))\end{aligned}$$

There is also one quick definition that we should get out of the way so that we can use it when we need to.

A **smooth curve** is any curve for which $\vec{r}'(t)$ is continuous and $\vec{r}'(t) \neq 0$ for any t except possibly at the endpoints. A helix is a smooth curve, for example.

Finally, we need to discuss integrals of vector functions. Using both limits and derivatives as a guide it shouldn't be too surprising that we also have the following for integration for indefinite integrals

$$\begin{aligned}\int \vec{r}(t) dt &= \left\langle \int f(t) dt, \int g(t) dt, \int h(t) dt \right\rangle + \vec{c} \\ \int \vec{r}(t) dt &= \int f(t) dt \vec{i} + \int g(t) dt \vec{j} + \int h(t) dt \vec{k} + \vec{c}\end{aligned}$$

and the following for definite integrals.

$$\begin{aligned}\int_a^b \vec{r}(t) dt &= \left\langle \int_a^b f(t) dt, \int_a^b g(t) dt, \int_a^b h(t) dt \right\rangle \\ \int_a^b \vec{r}(t) dt &= \int_a^b f(t) dt \vec{i} + \int_a^b g(t) dt \vec{j} + \int_a^b h(t) dt \vec{k}\end{aligned}$$

With the indefinite integrals we put in a constant of integration to make sure that it was clear that the constant in this case needs to be a vector instead of a regular constant.

Also, for the definite integrals we will sometimes write it as follows,

$$\int_a^b \vec{r}(t) dt = \left(\left\langle \int f(t) dt, \int g(t) dt, \int h(t) dt \right\rangle \right) \Big|_a^b$$
$$\int_a^b \vec{r}(t) dt = \left(\int f(t) dt \vec{i} + \int g(t) dt \vec{j} + \int h(t) dt \vec{k} \right) \Big|_a^b$$

In other words, we will do the indefinite integral and then do the evaluation of the vector as a whole instead of on a component by component basis.

Example 3 Compute $\int \vec{r}(t) dt$ for $\vec{r}(t) = \langle \sin(t), 6, 4t \rangle$.

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All we need to do is integrate each of the components and be done with it.

$$\int \vec{r}(t) dt = \langle -\cos(t), 6t, 2t^2 \rangle + \vec{c}$$

Example 4 Compute $\int_0^1 \vec{r}(t) dt$ for $\vec{r}(t) = \langle \sin(t), 6, 4t \rangle$.

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In this case all that we need to do is reuse the result from the previous example and then do the evaluation.

$$\begin{aligned} \int_0^1 \vec{r}(t) dt &= \left(\langle -\cos(t), 6t, 2t^2 \rangle \right)_0^1 \\ &= \langle -\cos(1), 6, 2 \rangle - \langle -1, 0, 0 \rangle \\ &= \langle 1 - \cos(1), 6, 2 \rangle \end{aligned}$$